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Re-evaluating Continual VQA: Toward Fair and Robust Evaluation for Multimodal Continual Learning

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Abstract

Continual Visual Question Answering (Continual VQA) poses unique challenges for multimodal continual learning, requiring models to incrementally acquire new knowledge while preserving visual–semantic grounding across tasks. However, existing benchmarks hinder fair and robust evaluation of such capabilities, as they allow models to exploit dataset biases rather than demonstrate genuine continual reasoning. We identify two structural flaws in current benchmark design. First, shared answer vocabularies across tasks encourage answer memorization, inflating performance and underestimating forgetting. Second, static answer priors within each task make the training and test answer distributions nearly identical, obscuring robustness under distribution shifts. To address these issues, we introduce UCo-VQA, an Unbiased benchmark suite that enforces token-level disjoint answer spaces across tasks and introduces intra-task train–test distribution shifts, enabling fairer assessment of forgetting and generalization in multimodal continual learning. We further provide a parameter-efficient baseline that mitigates forgetting and enhances grounding through question-only replay and dual-level distillation, offering a lightweight and memory-efficient framework for continual adaptation. Extensive experiments on UCo-VQA reveal that prior methods substantially overestimate performance under biased setups, while our approach achieves state-of-the-art results, improving robustness and retention by up to 4.18% and 2.21%, respectively[1].

1. Introduction

Recently, large pre-trained models have achieved remarkable success across language, vision, and multimodal domains [18, 46, 53]. In particular, vision-language (VL) models [26, 34, 47] demonstrate strong zero-shot generalization across a range of multimodal tasks. However, most of these models adopt a conventional training paradigm that assumes access to a fixed dataset available in full [10, 11, 19, 22, 23, 39, 40], overlooking the dynamic and evolving nature of real-world data. To address this limitation, Continual Learning (CL) [7, 13, 15–17, 55, 62, 65, 66] provides a lifelong learning framework that enables pre-trained VL models to acquire new knowledge incrementally without retraining from scratch. This capability is critical for deploying VL models in open-ended environments where input distributions evolve continuously.

Due to distribution shifts, continual adaptation often results in performance degradation on previously learned tasks, a phenomenon known as catastrophic forgetting [20, 43, 44]. The central challenge in CL is to acquire new knowledge without erasing prior knowledge [55]. While most research has focused on unimodal tasks such as image classification [13, 54, 58], the more complex yet practically important multimodal setting, particularly continual visual question answering (VQA), remains underexplored. Unlike unimodal tasks, continual VQA requires models to ground answers in both visual content and textual questions, and to preserve this grounding across evolving tasks.

While existing datasets such as VQA v2 [61] serve as standard benchmarks for evaluating continual VQA, they exhibit two key limitations. First, shared answer vocabularies across tasks can lead to spurious anti-forgetting performance (see Fig. 1). Because answers from earlier tasks (e.g., *yes*/*no*) often recur in later ones, models may appear to retain prior knowledge simply by matching repeated answers, rather than through genuine visual-semantic understanding. Second, these benchmarks overlook robustness evaluation under distribution shift. When the test distribution changes within a task, pre-trained models exhibit significant knowl-

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[1]Link to code-https://github.com/Zi-Jian-Gao/MaDQ

edge fragility (see Fig. 2). In CL settings, such fragility accumulates over time, leading to compounded forgetting and degraded generalization.

To address these limitations, we introduce **UCo-VQA**, a new benchmark suite based on VQA v2 [1, 61] and GQA [25, 27], each redesigned with two distinct CL splits. UCo-VQA features: (1) token-level disjoint answer spaces constructed by assigning task-specific answer vocabularies, and (2) answer distribution shifts within tasks to evaluate model robustness. Furthermore, we propose **MaDQ** (Matching and Distillation with Question replay), a continual VQA method that avoids storing complete image–question–answer triplets by replaying only past questions, reducing memory usage and preserving data privacy. MaDQ incorporates a dual-head architecture with answer prediction and image–question matching objectives, further regularized by temporal distillation. This is implemented through a parameter-efficient framework, where a working adapter learns new tasks, and a momentum adapter provides stable supervision for retaining prior knowledge.

Our main contributions are as follows: (1) We identify two key flaws in existing continual VQA benchmarks—shared answer vocabularies across tasks and static answer distributions within tasks—and introduce UCo-VQA, a fair and robust benchmark for evaluating forgetting and generalization. (2) We propose MaDQ, a lightweight and memory-efficient framework using question-only replay and dual-level distillation to enhance knowledge retention and grounding. (3) Experiments on UCo-VQA show that prior methods overestimate performance under biased setups, while MaDQ achieves new baselines and reveals a strong link between robustness and forgetting.

2. Related Work

Visual Question Answering. VQA requires models to reason over both visual and textual inputs. Recent advances in pre-trained VL models have led to strong performance on VQA benchmarks [34, 49]. However, studies have shown that existing VQA models often exploit superficial linguistic cues instead of grounding answers in visual content [5, 6, 29, 41], which limits their robustness especially under distribution shifts [2, 36]. To better assess model robustness, VQA-CP [1] and GQA-OOD [27] restructure the VQA v2 [21] and GQA [25] datasets by altering answer distributions between training and test splits. Nevertheless, these benchmarks are designed for static evaluations and do not consider robustness under CL scenarios. In this work, we demonstrate that robustness and anti-forgetting are closely related: *models with better visual grounding generalize more reliably across tasks*. We systematically investigate both forgetting and robustness in a continual setting by introducing new benchmarks and methods tailored to this challenge.

Multimodal Continual Learning. Continual learning (CL) aims to train a single model to incrementally acquire new knowledge while retaining prior tasks [55, 59, 63, 64]. Most existing CL research has focused on unimodal image classification [12, 38, 51, 56, 57], leaving sequential multimodal learning largely underexplored. Recently, several works have extended CL to vision–language scenarios [4, 30, 32, 42, 45, 52]. For example, RATT [9] explores continual learning for image captioning, while ConStruct-VL [50] and ZAF [14] study incremental concept-level reasoning. In the VQA domain, VQACL [61] introduces the first continual VQA benchmark based on pre-trained unimodal models and proposes a prototype-based strategy to mitigate forgetting. However, we identify two key limitations in VQACL: (1) repeated answer vocabularies lead to underestimated forgetting; and (2) it fails to evaluate robustness under distribution shifts. Our benchmark addresses both issues by enforcing token-level disjoint answer spaces across tasks and introducing train–test shifts in answer distributions.

3. A Fair and Robust Benchmark for Continual VQA

This section identifies two key structural flaws in existing continual VQA evaluation protocols and outlines our benchmark design that rectifies them.

3.1. Token-level Disjoint Answer Spaces for Fair Evaluation of Forgetting

Continual VQA [61] has recently emerged as a setting for evaluating continual learning across a sequence of evolving visual–question answering skills (e.g., *Location*, *Color*, *Count*). However, we identify a fundamental flaw in existing evaluation protocols: **spurious anti-forgetting performance**. For example, a model trained on the *Location* task may appear to retain high accuracy on previous tasks—not because of genuine knowledge retention, but due to **shared answer vocabularies** with tasks such as *Recognition*. When multiple tasks reuse common answers (e.g., "yes/no", "2", "red"), the model can rely on memorized answer priors rather than task-specific reasoning, leading to a systematic overestimation of its anti-forgetting capability.

To quantify this effect, we compute inter-task answer distribution overlap using skewed Kullback–Leibler (KL) divergence [31, 45]. Let p and q denote two answer distributions. Since zero-probability entries in q may cause instability in KL divergence, we adopt a smoothed mixture distribution and compute $-D_{\mathrm{KL}}(p \,\|\, (1-\alpha)p + \alpha q)$, where $\alpha = 0.99$. Higher values indicate greater overlap. We obtain an answer overlap matrix S, where each entry S_{ij} measures the degree of overlap between the training distribution of task $\mathcal{T}^i$ and the test distribution of task $\mathcal{T}^j$. In parallel, we construct a performance matrix A, where A_{ij} denotes the model's accuracy on task $\mathcal{T}^j$ after sequential training up to task $\mathcal{T}^i$ using a given CL method.

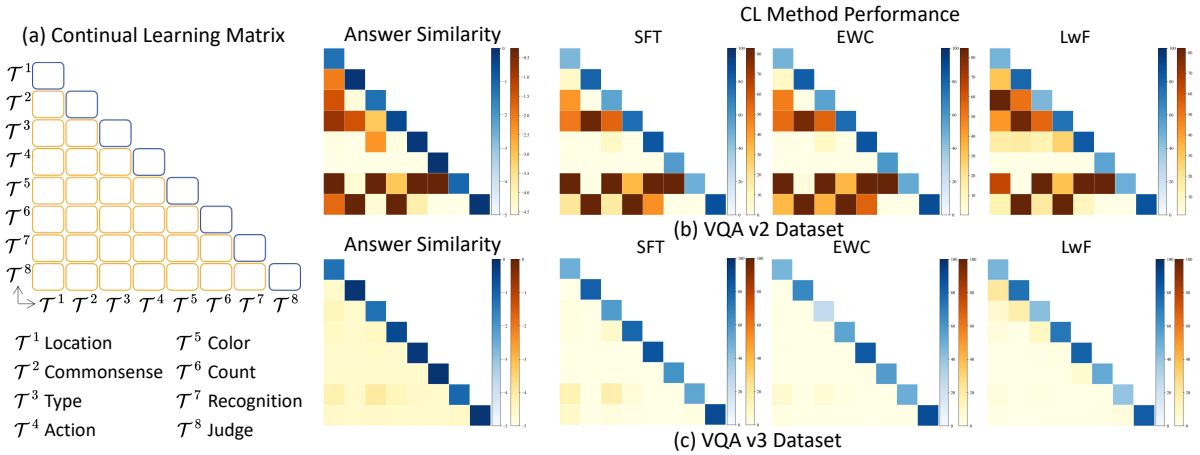


Figure 1. (a) Illustration of the continual learning (CL) evaluation matrix, where each cell represents a metric (e.g., answer similarity or accuracy) between task pairs, and the i-th row shows evaluations on all seen tasks after training up to $\mathcal{T}_i$. (b–c) Visualization of inter-task answer similarity and CL accuracy on VQA v2 and VQA v3 using three representative methods (SFT, EWC, and LwF). Higher inter-task similarity corresponds to smaller forgetting and inflated overall performance, revealing bias in existing benchmarks.

Fig. 1(b) shows the answer overlap and CL performance heatmaps on the VQA v2 dataset [61](see Appendix for additional datasets), using sequential fine-tuning (SFT) [24], EWC [28] and LwF [37]. The performance matrix shows notable fluctuations: even without any memory mechanism, SFT sometimes appears to recover or even exceed earlier-task performance after training on unrelated tasks. For example, the *Location* task swings as follows: 38.44 $\rightarrow$ 1.28 $\rightarrow$ 34.13 $\rightarrow$ 27.08 $\rightarrow$ 0.34 $\rightarrow$ 0.07 $\rightarrow$ 41.58 $\rightarrow$ 18.07. These shifts do not reflect genuine retention; they arise from *answer overlap* with subsequent tasks. For instance, a location query such as "Where is the kite?" may have the answer *left*, while in a *recognition* task the question "Which man is holding a beer?" may also yield *left*, producing spurious gains from shared labels rather than genuine knowledge retention. This effect is particularly pronounced for weaker baselines such as SFT. A clear correspondence between the similarity matrix S and the accuracy matrix A is visible and confirmed by Spearman's rank correlation on the flattened matrices (corr $= 0.73$ and 0.69, $p < 0.001$ for SFT and EWC; corr $= 0.42$, $p < 0.05$ for LwF). Thus, under standard evaluation, much of the apparent knowledge retention is attributable to inter-task answer sharing. The lower correlation for LwF indicates that its anti-forgetting strategies slightly suppress this effect; nevertheless, the evaluation remains confounded by residual answer sharing.

To address this issue, we redesign the VQA v2 dataset with **disjoint answer distributions across tasks**, resulting in a new variant we refer to as **VQA v3**. Overlaps in the original dataset are concentrated in binary responses (e.g., *yes/no*), frequent terms, and numeric values. Specifically, for binary questions (e.g., *yes/no*), we assign task-specific three-digit octal codes (e.g., 000/001 for $\mathcal{T}^1$, 002/003 for $\mathcal{T}^2$).[2] For open-ended answers, we adopt a dynamic prefix transformation strategy that systematically varies answer templates across tasks, e.g., from "it's [answer]" to "that's [answer]" and "there's [answer]". *This reconstruction preserves the semantic content of the original answers while removing lexical overlap, thereby enforcing task-specific separation at the token level.* Fig. 1(c) presents the corresponding heatmaps on our VQA v3 dataset. As expected, the spurious performance retention observed in Fig. 1(c) is no longer present, and the model exhibits a more consistent degradation as training progresses. These results validate our redesigned setting as a more unbiased and reliable benchmark for evaluating anti-forgetting.

3.2. Answer Distribution Shifts for Robustness Evaluation

While Sec. 3.1 addresses answer sharing across tasks, another critical yet underexplored challenge in continual VQA lies in the robustness of learned knowledge under distributional shifts. In the current setting of VQA v3, although answer distributions vary across tasks, the training and test sets within each task share identical answer distributions, a configuration we refer to as **standard splits (SS)**.

Prior work [41] has shown that VQA models often rely on superficial correlations between question patterns and frequent answers, rather than grounded visual-linguistic reasoning. For instance, a model may associate the question prefix "what color" with the answer "white" or "how many" with "2" regardless of image content. This shortcut behavior leads to brittle knowledge representations that perform well under the SS setting, but break down under distribution shifts in answer priors. To promote the development of continual VQA models with stronger generalization capabil-

[2]Three-digit octal codes provide up to 512 unique answer tokens while avoiding collisions with numeric values used in tasks like *Count*.

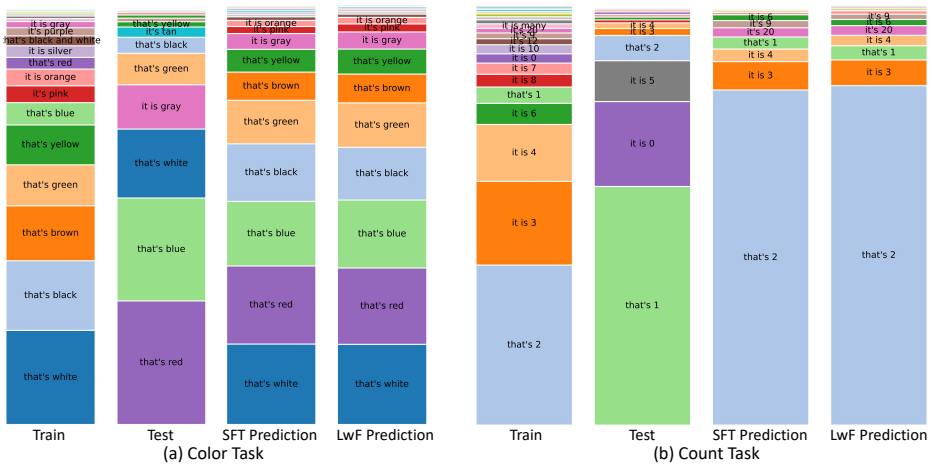


Figure 2. Train/test answer distributions and model predictions under the proposed VQA v3 splits (PS). (a) *Color* and (b) *Count* tasks. Each column shows the answer distribution in the training set, test set, and model predictions (SFT and LwF). Models largely follow the training answer distribution, indicating limited robustness to distributional shifts.

Table 1. Overview of the **UCo-VQA Benchmark Suite**. 'SS' denotes standard splits, and 'PS' denotes proposed splits with distributional shifts. Answer vocabularies may share tokens across tasks (shared) or be token-level disjoint.

Dataset	Split Type	Answer Space	Remark
VQA v2	SS	Shared	Original benchmark
VQA v3	**SS / PS**	**Disjoint**	**Proposed debiased version**
GQA	SS	Shared	Original benchmark
GQA v2	**SS / PS**	**Disjoint**	**Proposed debiased version**

ities, we introduce a new setting of **proposed splits (PS)**. In PS, the training and test sets for each task in VQA v3 are constructed to have distinct answer distributions, creating controlled intra-task distribution shifts.

To assess model robustness under these conditions, we evaluate SFT and LwF on the *Color* and *Count* tasks, comparing the answer distributions in training and test sets, as well as the models' output distributions on the test sets. As shown in Fig. 2, both models produce predictions that closely align with the training distribution rather than the ground-truth distribution in the test set, highlighting a lack of robust generalization and semantic grounding.

Following the same procedure as in VQA v2 to v3, we adapt GQA [25] into **GQA v2**, incorporating token-level disjoint answer spaces under both standard (SS) and proposed (PS) splits. Together with **VQA v3**, these form the **UCo-VQA** benchmark suite (see Tab. 1), which enables fair evaluation of both anti-forgetting and robust generalization under task-level distributional shifts. By eliminating answer sharing across tasks and introducing train–test distribution shifts within tasks, UCo-VQA provides a more reliable and unbiased testbed for multimodal continual learning.

4. Methodology

4.1. Background and Notation

Pre-trained Vision-Language Models. Pre-training provides a strong foundation for continual learning by both improving downstream task performance and mitigating catastrophic forgetting. The effectiveness of a pre-trained model largely depends on its representational capacity, which is influenced by data quality, model architecture, and the alignment between pre-training objectives and downstream tasks. Generally, better data quality, more expressive architectures, and higher task alignment lead to stronger generalizability.

In this work, we adopt the Bootstrapped Language Image Pre-training (BLIP) model [34], augmented with captioning and filtering strategies. BLIP is chosen for its strong multimodal representation capabilities, and importantly, its lack of VQA-specific fine-tuning ensures reliable and unbiased evaluation in continual VQA settings. The model architecture comprises four main components: an image encoder f_ν for extracting visual features, a question encoder f_τ for cross-modal fusion, an answer decoder f_ω for generating responses, and a classification head h_{cls} for final answer prediction. Given an input image x and a question q, the prediction process is defined as:

$$\phi(x, q) = h_{\text{cls}}(f_\omega(f_\tau(f_\nu(x), q))). \tag{1}$$

Continual VQA. We consider a continual VQA setting where a pre-trained VL model is sequentially adapted to a stream of VQA tasks, denoted as $\{\mathcal{T}^1, \mathcal{T}^2, ..., \mathcal{T}^n\}$. Each task $\mathcal{T}^t$ consists of a set of VQA samples, represented as triplets (x^t, q^t, a^t),[3] where $x^t \in \mathcal{X}^t$ is the image, $q^t \in \mathcal{Q}^t$ is a natural language question, and $a^t \in \mathcal{A}^t$ is the corresponding answer. Each answer is drawn from a task-specific

[3]The sample index within each task is omitted for clarity.

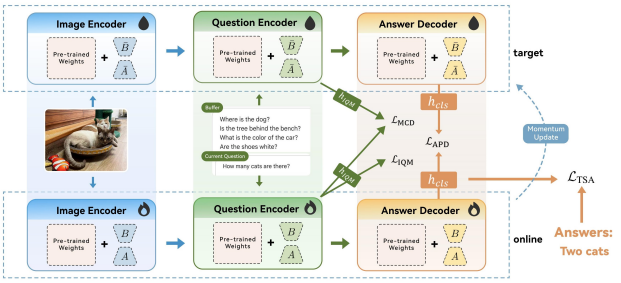


Figure 3. Training pipeline of the proposed method with question replay and dual-level distillation. The *lower* row shows the trainable working adapters for task adaptation, while the *upper* row depicts the momentum adapters serving as stable references for distillation.

answer space, as defined in Sec. 3.1. Under the PS setting, the answer distribution in the test set differs from that of the training set within each task, whereas the SS setting assumes the two distributions are identical (see Sec. 3.2). Unlike classification-based VQA formulations that select an answer from a predefined set, we adopt a *generative* framework [41, 61], where the model is expected to generate the correct answer a^t conditioned on both x^t and q^t. This is more practical and flexible, particularly for open-ended reasoning. In this work, each task $\mathcal{T}^t$ is designed to emphasize a distinct cognitive skill. During inference, traditional CL methods that require task identity become impractical due to the ambiguity in determining task boundaries from free-form natural language. As such, when learning a new task $\mathcal{T}^t$, the model must acquire the new skill while maintaining performance across all previously learned tasks $\{\mathcal{T}^1, \ldots, \mathcal{T}^{t-1}\}$.

4.2. Continual VQA Objective

To support effective continual learning in VQA, we propose **MaDQ** (Matching and Distillation with Question replay), a method designed to address two fundamental challenges: (i) learning new tasks while preserving knowledge from previous ones under inter-task distribution shifts; and (ii) improving robustness to intra-task distribution shifts, as posed by the PS setting (Sec. 3.2). As illustrated in Fig. 3, MaDQ achieves this by integrating four hierarchical components into its training objective: task-specific answering ($\mathcal{L}_{\text{TSA}}$), answer prediction distillation ($\mathcal{L}_{\text{APD}}$), image–question matching ($\mathcal{L}_{\text{IQM}}$), and matching consistency distillation ($\mathcal{L}_{\text{MCD}}$):

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{TSA}}}_{\text{learning}} + \underbrace{\mathcal{L}_{\text{APD}}}_{\text{retention}} + \underbrace{(\mathcal{L}_{\text{IQM}} + \mathcal{L}_{\text{MCD}})}_{\text{robustness}}. \quad (2)$$

Task-Specific Answering ($\mathcal{L}_{\text{TSA}}$). To acquire new task-specific knowledge, the CL model is trained on the current task $\mathcal{T}^t$ using the standard cross-entropy loss:

$$\mathcal{L}_{\text{TSA}} = \frac{1}{|\mathcal{T}^t|} \sum_{(x^t, q^t, a^t) \in \mathcal{T}^t} \mathcal{L}_{\text{CE}}(\phi^t(x^t, q^t), a^t), \quad (3)$$

where $\phi^t(x^t, q^t)$ denotes the answer distribution predicted by the current model for the current image–question pair.

Answer Prediction Distillation ($\mathcal{L}_{\text{APD}}$). To mitigate forgetting of previous tasks while avoiding the storage of visual data, MaDQ adopts a memory-efficient *question-only* replay strategy. We maintain a buffer $\mathcal{M}$ containing questions from previous tasks and pair them with current task images $\mathcal{X}^t$ to construct pseudo-samples (x^t, q^i). Although these image–question pairs may be semantically misaligned, replayed questions serve as useful cues for recalling task-specific knowledge in continual VQA. To preserve previously acquired knowledge, we enforce KL-based consistency on answer predictions from pseudo-samples across tasks:

$$\mathcal{L}_{\text{APD}} = \frac{1}{|\mathcal{X}^t||\mathcal{M}|} \sum_{x^t \in \mathcal{X}^t, q^i \in \mathcal{M}} \mathcal{L}_{\text{KL}}(\phi^t(x^t, q^i), \phi^{t-1}(x^t, q^i)), \quad (4)$$

where $\phi^t(x^t, q^i)$ and $\phi^{t-1}(x^t, q^i)$ denote the answer distributions predicted by the current and previous models, respectively, for the same pseudo image–question pair.

Image-Question Matching ($\mathcal{L}_{\text{IQM}}$). As discussed in Sec. 3.2, VQA models often rely on superficial linguistic correlations during CL, which weakens their visual grounding. This shortcut behavior leads to brittle reasoning and degrades performance on both current and previous tasks, particularly under the PS setting, where test-time linguistic patterns differ from those seen during training. To mitigate this issue, we introduce a binary classification head h_{IQM} that predicts whether an image–question pair is semantically matched. This component explicitly encourages the model to align visual and linguistic inputs. Positive pairs (x^t, q^t) are drawn from the current task $\mathcal{T}^t$, while negative pairs (x^t, q^i) are constructed by pairing current task images with replayed questions from the buffer $\mathcal{M}$. Using the same encoders and decoder in Eq. (1), the matching probability is computed as:

$$\psi^t(x^t, q^j) = h_{\text{IQM}}(f_\tau(f_\nu(x^t), q^j)), \quad \text{where} \quad q^j \in \mathcal{Q}^t \cup \mathcal{M}, \quad (5)$$

and the associated loss is given by:

$$\mathcal{L}_{\text{IQM}} = \frac{1}{|\mathcal{X}^t|\,|\mathcal{Q}^t \cup \mathcal{M}|} \sum_{x^t \in \mathcal{X}^t,\ q^j \in \mathcal{Q}^t \cup \mathcal{M}} \mathcal{L}_{\text{CE}}(\psi^t(x^t, q^j), y), \quad (6)$$

where $y \in \{0, 1\}$ indicates whether the image–question pair is a semantic match. This objective reinforces cross-modal grounding [33, 34] and enhances robustness against both modality bias and distributional shifts.

Matching Consistency Distillation ($\mathcal{L}_{\text{MCD}}$). In addition to the answer-level regularization in Eq. (4), we enforce consistency in image–question matching behavior (Eq. (7)) across tasks, thereby preserving both negative rejection boundaries and newly acquired positive associations:

$$\mathcal{L}_{\text{MCD}} = \frac{1}{|\mathcal{X}^t||\mathcal{Q}^t \cup \mathcal{M}|} \sum_{x^t \in \mathcal{X}^t, q^j \in \mathcal{Q}^t \cup \mathcal{M}} \mathcal{L}_{\text{KL}}(\psi^t(x^t, q^j), \psi^{t-1}(x^t, q^j)), \quad (7)$$

where $\psi^t(x^t, q^j)$ and $\psi^{t-1}(x^t, q^j)$ denote the matching probabilities predicted by the current and previous models. Together, $\mathcal{L}_{\text{IQM}}$ and $\mathcal{L}_{\text{MCD}}$ jointly act on both current and replayed questions to promote robust generalization and stable vision–language grounding in multimodal CL.

4.3. Dual-LoRA Architecture

To enable efficient and stable continual learning, we propose a **Dual-LoRA architecture**, which integrates low-rank adapters into the vision-language backbone and maintains two distinct LoRA branches: a trainable working adapter (**LoRA-w**) for task adaptation, and a momentum adapter (**LoRA-m**) for distillation reference. This design avoids full fine-tuning of the pre-trained model while providing stable access to past predictions required for knowledge retention.

Structure. As depicted in Fig. 3, we inject LoRA adapters into all layers of the image encoder f_ν, question encoder f_τ, and answer decoder f_ω, including the query/key/value (Q/K/V) projections, feed-forward layers, and token embedding layers. Given a frozen pre-trained weight W^0, each trainable weight W is reparameterized as $W = W^0 + BA$, where $A \in \mathbb{R}^{r\times k}$ and $B \in \mathbb{R}^{d\times r}$ are learnable low-rank matrices, with $r \ll \min(d, k)$. In addition to the backbone, the answer prediction head h_{cls} and the matching head h_{IQM} are appended on top of the encoder/decoder outputs. These heads are fixed after the first task to avoid overfitting. A detailed training procedure for MaDQ is provided in Algorithm 1 of the Appendix.

Learning and Inference. During each task $\mathcal{T}^t$, only the LoRA-w adapters $\{A, B\}$ are optimized, and probabilities ϕ^t, ψ^t are predicted using $W = W^0 + BA$. After each epoch, we update the LoRA-m adapters $\{\bar{A}, \bar{B}\}$ using an exponential moving average (EMA):

$$\bar{A} \leftarrow \alpha\bar{A} + (1-\alpha)A, \quad \text{and} \quad \bar{B} \leftarrow \alpha\bar{B} + (1-\alpha)B. \tag{8}$$

The momentum update produces a stable reference model, $W^t = W^0 + \bar{B}\bar{A}$, which serves as a teacher for both answer-prediction distillation (Eq. (4)) and matching-consistency distillation (Eq. (7)), as well as for inference.

5. Experiments

5.1. Experimental Setup

Benchmarks. We evaluate all methods on the proposed UCo-VQA, which includes two datasets (VQA v3 and GQA v2), each evaluated under two settings: Standard Split (SS) and Proposed Split (PS), as described in Sec. 3. In both settings, we enforce token-level disjoint answer spaces across tasks to better measure true forgetting. In addition, the PS setting introduces a distribution shift between training and testing phases within each task, allowing evaluation of robustness under realistic answer distribution changes. The VQA v3 benchmark comprises an 8-task sequence: `location` → `commonsense` → `type` → `action` → `color` → `count` → `recognition` → `judge`. GQA v2 includes 7 tasks: `action` → `material` → `logical` → `object` → `color` → `position` → `relation`. Each question in GQA v2 has a single answer, while VQA v3 includes multiple human-annotated answers per question.

Compared Methods. We compare MaDQ with a broad range of continual learning baselines, covering both replay-free and replay-based paradigms. For methods without replay, we include EWC [28] and LwF [37], as well as multimodal approaches like Layered-LoRA [50], MoE-Adapters [60], and ZAF [14]. For replay-based methods, we consider Experience Replay (ER) [48] and CLS-ER [3] which incorporates biologically inspired consolidation and distillation strategies. We also compare against GAB [8], a continual VQA method that additionally fine-tunes a large vision-language model on each new task to generate question-answer pairs for previous task types, conditioned on images from the new task. Additionally, we include two reference setups: (1) *Joint Learning*, where all tasks are trained simultaneously as an upper bound, and (2) *Sequential Fine-tuning (SFT)*, where each task is trained without any forgetting mitigation, representing a lower bound. Both are implemented with LoRA [24] for consistent parameterization. *We do not include VQACL [61] in our comparison, since it targets a dual-label formulation that jointly predicts question type and object category. Its prototype-based design requires object-level supervision, which is unavailable in our benchmark and thus not directly comparable.*

Implementation Details. To ensure fairness, all methods are implemented using the same BLIP backbone [34], initialized with the CapFilt-enhanced pre-trained weights. Following prior work [14, 50], we use a weight decay of 0.05, an initial learning rate of 1.25e-3 with a cosine scheduler, and train for 12 epochs per task. All low-rank adapters are configured with rank $r = 16$, and the momentum coefficient α is initialized to 0.85. Additional implementation and dataset construction details are included in the Appendix.

Evaluation Metrics. We report three standard continual learning metrics [55]: Final Average Accuracy (FAA), Cumulative Average Accuracy (CAA), and Final Forgetting Measure (FFM). Let A_{ij} denote the accuracy on task $\mathcal{T}^j$ after training on task $\mathcal{T}^i$. The average accuracy after task $\mathcal{T}^i$ is defined as $AA_i = \frac{1}{i}\sum_{j=1}^{i} A_{ij}$. We compute **FAA** as AA_n, i.e., the final average accuracy after all tasks; **CAA** as $\frac{1}{n}\sum_{i=1}^{n} AA_i$, reflecting average performance throughout training; and **FFM** as $\frac{1}{n-1}\sum_{j=1}^{n-1}(A_{jj} - A_{nj})$, which measures the average forgetting on previous tasks after learning the final task. FAA highlights the final performance gap to Joint Learning, CAA captures the overall learning quality, and FFM directly quantifies forgetting.

5.2. Results and Analysis

Overall Performance. As shown in Tab. 2, MaDQ consistently achieves the best performance across all four benchmark settings, attaining the highest FAA and CAA scores while maintaining the lowest FFM, indicating strong generalization and resistance to forgetting. Parameter-efficient continual learning baselines such as Layered-LoRA and MoE-

Table 2. Comparison of CL methods on VQA v3 and GQA v2 under SS and PS settings. MaDQ achieves the best overall performance across both settings, showing consistent gains in FAA and CAA and reduced forgetting, particularly under the more challenging PS setting.

Method	#Memory Per Task	Memory Type	Storage Usage (MB)	VQA v3 Standard Splits (SS)			VQA v3 Proposed Splits (PS)		
				FAA(↑)	*CAA*(↑)	*FFM*(↓)	*FAA*(↑)	*CAA*(↑)	*FFM*(↓)
Joint Learning	0	–	–	69.41	-	-	42.90	-	-
SFT (ICLR 2021) [24]	0	–	–	13.05	23.25	64.98	7.41	14.74	37.99
Layered-LoRA (CVPR 2023) [50]	0	–	–	13.21	23.19	63.23	7.55	14.88	37.07
MoE-Adapters (CVPR 2024) [60]	0	–	–	11.23	19.63	59.58	6.00	12.57	36.13
EWC (PNAS 2017) [28]	0	–	–	11.45	18.70	54.59	6.91	12.80	35.84
LwF (TPAMI 2018) [37]	0	–	–	13.33	25.44	55.41	7.58	11.85	29.28
ZAF (NeurIPS 2024) [14]	0	–	–	24.80	35.99	47.77	11.88	25.08	32.95
GAB (WACV 2025) [8]	0	–	–	26.54	31.69	46.99	13.15	27.94	29.94
ER (NeurIPS 2019) [48]	400	? / 🖼	67.53	52.48	54.31	18.11	32.24	33.64	9.27
CLS-ER (ICLR 2022) [3]	400	? / 🖼	67.53	52.54	54.53	17.45	34.32	32.58	7.56
MaDQ (Ours)	400	?	0.01	**53.21**	**57.62**	**16.79**	**36.15**	**37.82**	**4.54**
Improvement	–	–	–	***0.67***	***3.09***	***0.66***	***1.83***	***4.18***	***3.02***

Method	#Memory Per Task	Memory Type	Storage Usage (MB)	GQA v2 Standard Splits (SS)			GQA v2 Proposed Splits (PS)		
				FAA(↑)	*CAA*(↑)	*FFM*(↓)	*FAA*(↑)	*CAA*(↑)	*FFM*(↓)
Joint Learning	0	–	–	45.70	-	-	44.93	-	-
SFT (ICLR 2021) [24]	0	–	–	6.43	16.92	50.79	5.48	15.88	45.63
Layered-LoRA (CVPR 2023) [50]	0	–	–	6.31	16.79	50.64	5.43	15.71	45.26
MoE-Adapters (CVPR 2024) [60]	0	–	–	6.10	16.02	48.73	5.66	15.40	45.16
EWC (PNAS 2017) [28]	0	–	–	4.11	14.94	42.82	4.94	12.37	33.79
LwF (TPAMI 2018) [37]	0	–	–	6.83	13.82	14.56	9.09	14.68	10.76
ZAF (NeurIPS 2024) [14]	0	–	–	10.04	20.64	37.22	11.29	21.76	34.11
GAB (WACV 2025) [8]	0	–	–	20.78	26.89	28.48	17.82	24.31	29.98
ER (NeurIPS 2019) [48]	300	? / 🖼	46.89	41.61	42.02	12.68	39.45	42.93	13.40
CLS-ER (ICLR 2022) [3]	300	? / 🖼	46.89	41.71	42.85	11.21	39.57	42.97	11.87
MaDQ (Ours)	300	?	0.01	**43.51**	**43.00**	**6.96**	**40.72**	**45.18**	**9.37**
Improvement	–	–	–	***1.80***	***0.15***	***4.25***	***1.15***	***2.21***	***1.39***

Adapters exhibit substantial forgetting due to the absence of dedicated forgetting mitigation mechanisms. Regularization-based approaches like EWC and LwF also underperform in multimodal settings, with LwF suffering from constrained adaptability due to its reliance on the last-task model. Although ZAF incorporates knowledge distillation from external data, its effectiveness is limited by domain mismatch between its wild data source and the downstream VQA tasks. Meanwhile, GAB's reliance on fine-tuning a large vision–language model with limited data makes it brittle: even minor distributional drift between the generated image–question pairs and the true task distribution leads to a sharp performance degradation. Instead, replay-based methods (ER and CLS-ER) *benefit from explicit memory but require full image–question–answer triplets*, raising concerns over privacy and memory cost. In contrast, MaDQ achieves stronger performance while replaying only past questions. Specifically, under our PS setting, MaDQ surpasses traditional replay-based methods by **4.18%** and **2.21%** in CAA on VQA v3 and GQA v2, respectively. These results highlight the effectiveness of MaDQ's dual-level distillation design and matching-based regularization. Moreover, under mismatched answer distributions, most methods exhibit noticeable forgetting, whereas MaDQ consistently achieves the lowest FFM. This highlights that *greater robustness to distributional shifts can effectively mitigate forgetting in continual VQA*. Additional analyses on task order variations and the momentum hyperparameter α are included in the Appendix.

As shown in Fig. 4, models trained on the original GQA benchmark exhibit substantially higher FAA and CAA

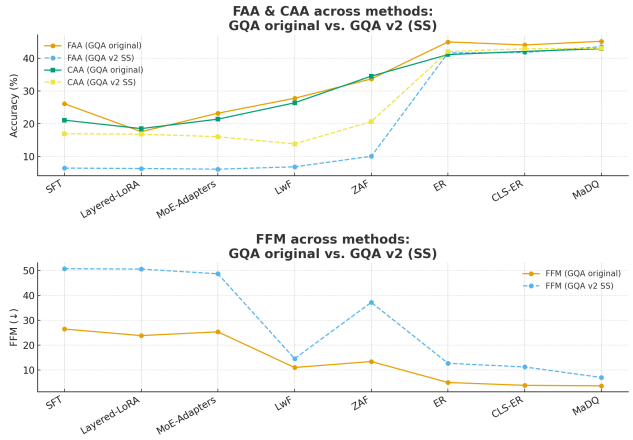


Figure 4. Comparison of CL methods on the original GQA and the debiased GQA v2 (SS). (**Top**) FAA and CAA; (**Bottom**) FFM.

scores, alongside lower FFM values. This apparent performance gain, however, stems largely from overlapping answer tokens across tasks, which allow models to exploit shared lexical patterns rather than maintain true task-specific knowledge. In contrast, the proposed GQA v2 (SS) enforces token-level disjoint answer spaces, resulting in a more realistic and challenging evaluation. Under this fairer setting, most methods experience a noticeable performance drop, revealing the extent of bias in existing benchmarks. Remarkably, MaDQ maintains comparable FAA/CAA and achieves consistently low FFM, demonstrating stronger knowledge retention and robustness to distributional shifts (see Appendix for results on VQA v2 and VQA v3 (SS)).

Table 3. Performance comparison on VQA v3 (Standard Splits) using the multi-modal large language model BLIP2 [35].

Metric	SFT	Layered-LoRA	LwF	ER	CLS-ER	MaDQ*
FAA (↑)	7.07	8.12	0.53	23.46	24.45	21.48
CAA (↑)	15.64	17.52	10.61	31.28	31.71	30.13
FFM (↓)	34.77	32.48	16.54	14.23	13.77	14.80

Table 4. Impact of each loss component in the proposed MaDQ.

$\mathcal{L}_{\text{APD}}$	$\mathcal{L}_{\text{IQM}}$	$\mathcal{L}_{\text{MCD}}$	GQA v2 (PS)			VQA v3 (PS)		
			FAA(↑)	*CAA*(↑)	*FFM*(↓)	*FAA*(↑)	*CAA*(↑)	*FFM*(↓)
			4.98	17.08	44.96	6.56	13.41	37.75
	✓		5.06	17.50	44.69	6.75	13.12	36.33
✓			36.88	41.75	14.29	31.81	32.74	10.34
✓	✓		40.22	43.36	11.07	35.50	36.92	6.83
✓	✓	✓	**40.72**	**45.18**	**9.37**	**36.15**	**37.82**	**4.54**

Table 5. Performance improvement from integrating $\mathcal{L}_{\text{IQM}}$ into existing CL methods on two benchmarks.

Method	GQA v2 (PS)			VQA v3 (PS)		
	FAA (↑)	CAA (↑)	FFM (↓)	**FAA** (↑)	CAA (↑)	FFM (↓)
LwF [37]	9.09	14.68	10.76	7.58	11.85	29.28
w/ $\mathcal{L}_{\text{IQM}}$	**9.63**	**14.70**	**8.82**	**9.27**	**14.22**	**28.10**
ER [48]	39.45	42.93	13.40	32.24	33.64	9.27
w/ $\mathcal{L}_{\text{IQM}}$	**40.39**	**43.58**	**11.13**	**33.65**	**35.18**	**8.50**
CLS-ER [3]	39.57	42.97	11.87	34.32	32.58	7.56
w/ $\mathcal{L}_{\text{IQM}}$	**40.97**	**44.57**	**10.71**	**35.44**	**34.43**	**5.46**

Meanwhile, Tab. 3 reports results when replacing BLIP with the multi-modal large language model BLIP2 [35] on VQA v3 (SS). To adapt to BLIP2's architecture with a frozen vision encoder and a general LLM decoder, we employ a simplified variant, MaDQ*, which performs only answer-prediction distillation with question-only replay. In contrast, ER stores full image–question–answer triplets, and CLS-ER further adds distillation. Across all methods, BLIP2-based models yield notably lower FAA and CAA compared to BLIP-based results in Tab. 2, indicating that large, generic LVLMs are less data-efficient and harder to adapt under our debiased continual VQA setting. This may be because BLIP2's LLM decoder is not optimized for short, structured answers and requires re-alignment with limited supervision. Nevertheless, MaDQ* matches ER and CLS-ER without storing ground-truth answers, validating the effectiveness and generality of our lightweight replay and distillation strategy.

Ablation Study. Tab. 4 analyzes the contribution of each objective in our MaDQ method on GQA v2 (PS) and VQA v3 (PS). Without distillation ($\mathcal{L}_{\text{APD}}$ and $\mathcal{L}_{\text{MCD}}$), performance remains near random levels (e.g., FAA = 5.06 on GQA v2), confirming the necessity of knowledge retention. Introducing $\mathcal{L}_{\text{APD}}$ alone substantially boosts performance (FAA +31.9 on GQA v2), while adding the image–question matching loss $\mathcal{L}_{\text{IQM}}$ further enhances both anti-forgetting and robustness (e.g., CAA +1.61). Finally, incorporating the matching-consistency distillation term $\mathcal{L}_{\text{MCD}}$ achieves the best overall performance, reducing forgetting (FFM to 9.37 on GQA v2 and 4.54 on VQA v3) and improving both FAA and CAA. These results demonstrate that $\mathcal{L}_{\text{IQM}}$ and $\mathcal{L}_{\text{MCD}}$ play complementary roles in preserving visual–semantic grounding and stabilizing continual learning.

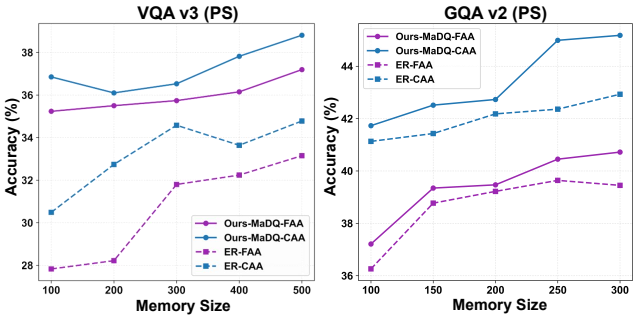


Figure 5. Performance comparison of MaDQ and ER under varying memory sizes per task.

Plugin Analysis. Tab. 5 shows the effect of integrating $\mathcal{L}_{\text{IQM}}$ into representative CL baselines. Across both benchmarks, this simple plug-in consistently reduces forgetting—by up to **2.33%** on GQA v2—and often improves FAA and CAA. Although originally designed to enhance robustness, $\mathcal{L}_{\text{IQM}}$ also yields notable gains in mitigating forgetting, suggesting that improved robustness facilitates better knowledge retention.

Memory Sensitivity Analysis. Fig. 5 compares MaDQ and ER under varying memory sizes per task. While ER stores complete image–question–answer triplets, MaDQ retains only past questions. Despite this lightweight design, MaDQ consistently achieves higher CAA and comparable FAA across both benchmarks. These results demonstrate that question-only replay, combined with our distillation strategy, is not only effective but also memory-efficient, highlighting MaDQ's scalability and practicality for real-world continual learning scenarios with limited storage capacity.

6. Discussion and Limitation

This work re-examines continual VQA from an evaluation perspective, revealing two structural flaws in existing benchmarks: shared answer vocabularies across tasks and static answer distributions within tasks. These issues lead to biased assessments of forgetting and robustness. To address them, we introduce UCo-VQA, a redesigned benchmark suite built upon VQA v2 and GQA, which enforces token-level disjoint answer spaces across tasks and introduces train–test distribution shifts within each task. On this foundation, we propose MaDQ, a parameter-efficient continual learning framework that replays only past questions and employs dual-level distillation to preserve prior knowledge and strengthen visual–semantic grounding. Experiments show that improving robustness through matching regularization also enhances forgetting resistance, underscoring the intrinsic link between grounding stability and continual generalization. While MaDQ minimizes memory overhead, it still relies on question replay, which may raise privacy concerns—an aspect worth exploring in future work on more open-ended or noisy continual scenarios.

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